

Clinical Comorbidity Dashboard

FP-Growth association rule mining over clinical visits

Streamlit | mlxtend | Dolt | Deployed at clinical-comorbidity-dashboard.streamlit.app

Problem & Objective

- > Hospitals capture rich diagnostic data, but co-occurrence patterns between conditions stay buried in tables.
- > Goal: surface clinically meaningful comorbidity patterns that support earlier screening and multi-disciplinary care planning.
- > Approach: mine 2,440 visits with FP-Growth + association rules and expose the rule set through an interactive Streamlit dashboard.
- > Audience: clinicians and care coordinators - not data scientists - so the UI must explain itself at a glance.

Dataset & Dolt Workflow

- > 2,440 patient encounters - primary diagnosis + secondary conditions, demographics, treatment markers.
- > Versioned with Dolt (Git-for-data): clone, sql -q export, commit, push - every rule set is reproducible against a commit hash.
- > data_raw/raw_data.csv -> transactions/transactions.csv -> data_processed/{frequent_itemsets, association_rules}.csv.
- > Streamlit app loads the processed CSVs at startup; the notebook rebuilds them from raw + Dolt for full reproducibility.

Why FP-Growth

- > Apriori generates exponentially many candidates - prohibitive at low support thresholds.
- > FP-Growth scans the data twice and builds a compressed FP-tree, then mines patterns recursively. ~3x faster on this dataset (0.4s vs 1.2s).
- > Parameters: min_support = 0.01, min_confidence = 0.50, lift > 1.
- > Output: 176 association rules retained after the lift filter.

Headline Results

176

Total rules

8.48

Max lift

78.3%

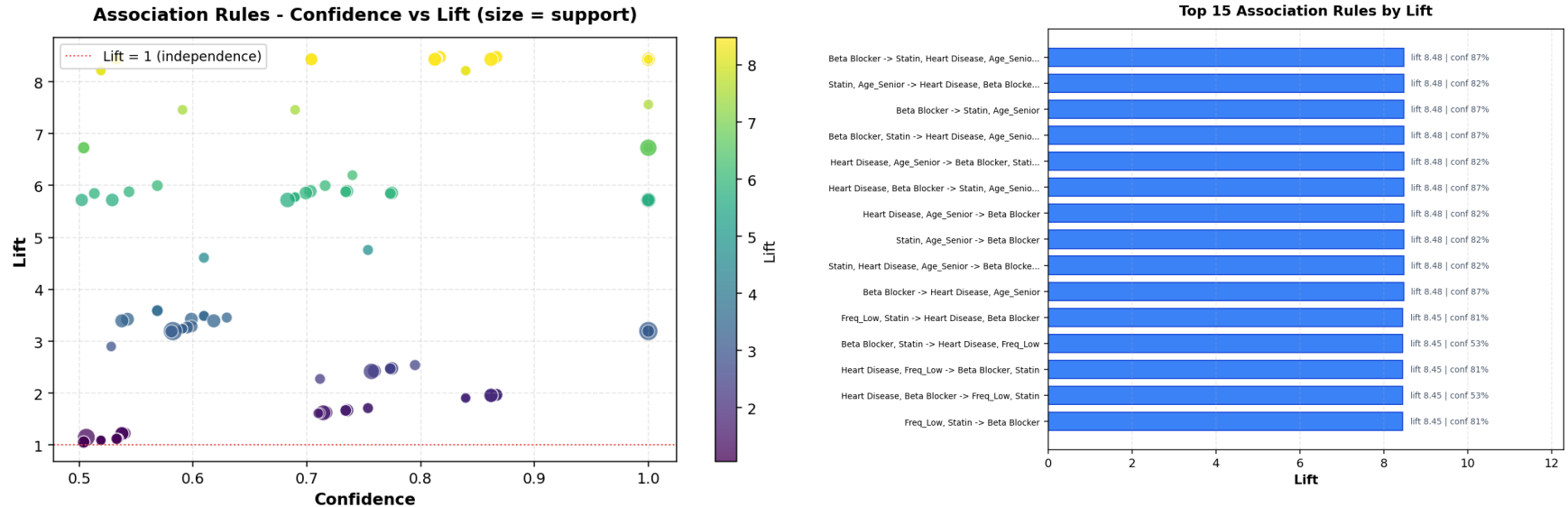
Avg confidence

2,440

Patients

Top rule by lift links Beta-Blocker therapy to a Statin + Heart Disease + Senior cohort - lift 8.48, confidence 86.6%. The dashboard ranks rules by lift by default; clinicians filter live by primary and secondary diagnosis.

Pattern Visualizations



Left: confidence vs lift, bubble size = support - actionable rules cluster top-right. Right: top 15 rules by lift, confidence annotated. Both regenerate from the same notebook cells.

Dashboard UI - Design Choices

- > Single-screen, 3-column layout - care plan / metric stack / vitals & heatmap. No tab nesting; everything visible at once.
- > @st.dialog modals for deep dives (Demographics, Multi-Disciplinary Consult, Patient #2440) - keep the home view uncluttered.
- > Animated 3D anatomical backdrop establishes domain context without dominating attention (0.55 opacity, 18s rotation).
- > Hardened CSS: forced dark text, gradient-safe overrides, and selectbox styling so every control reads on any display.
- > Reactive caching (@st.cache_data) - filter changes apply in <100 ms over 176 rules.

Recap & Links

- > Versioned data (Dolt) -> FP-Growth pipeline -> 176 rules -> interactive Streamlit dashboard - reproducible end-to-end.
- > Deliverables: notebook, 7-page report, this deck, live app.
- > Future work: temporal sequence mining, rule confidence intervals, patient-trajectory clustering.

Live App: <https://clinical-comorbidity-dashboard.streamlit.app>
Repo: <https://github.com/Alan-911/clinical-comorbidity-dashboard>
Notebook: notebooks/clinical_comorbidity_analysis.ipynb
Report: docs/Clinical_Comorbidity_Dashboard_Report.pdf